

FINAL REPORT ? AB-Cloud v23 SUPERCOMBO ? interactive session

Generated: 2026-09-16 01:48:55 | Julia 1.12.0 | wall duration: 93748.1 s (1562.5 min) | Suite: AB-Cloud v20

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What this folder is

FINAL_REPORT is the LAST folder of the run and gathers ALL run information: the master report (this document, in md/html/pdf/docx) ? which since v23.5 EMBEDS every test's full report

How to reproduce

Full suite: `julia ab_cloud_v20.jl --test all`

Single test: `julia ab_cloud_v20.jl --test 33 --no-two-pass`

Disorder cap: `--ab-w-max 1.0` (Tests 32/33/36 use $W_{\text{eff}} = \min(\text{ab_}W, W_{\text{max}})$)

Test 36 targets: `--ab37-n-eigs-pass1 25000 --ab37-n-eigs-pass2 50000`

Labs: main menu 'l' (Physics Lab), 'd3' (3D Lab); every experiment writes its own report into the same run folder.

zeros_count = 50000 (source: auto)

Configuration snapshot (end-of-run values)

code path: /home/isaev/ab-cloud

lattice secondary = 96x96 Nv=[2] (two-pass=true, primary-only=false)

third pass (SERIES) = true (Tests 19/29/31 ? 19c/29c/31c series; Test 19 also runs pass 4 ? 19d twist AUDIT)

alpha = ab_alpha=0.5, ab_alpha_test16=0.5

disorder = ab_W=4, ab_W_max=1 ? W_{eff} (Tests 32/33/36)=1.00

test33 comparison row = W_cmp=0.50, Nv_cmp=144

realizations = ab_n_realizations=5, center_fraction=0.6

byte robustness = tol=0.02, n_bytes=200000, seed=12345, T37 targets: pass1=25000 / pass2=50000 eigs

lab = L=72, ?=0.5, W=4, q=1, Nv=4, random/torus/monumental, seed=12345

3d = L=20, ?=0.5, tz=1, W=0, mass=0, boundary=:torus, lines=0 q=1, seed=777

reporting = enabled=true, dir=results, dpi=600, formats=md,html,pdf,docx

Test 1 [PASS] bN_convergence ? Test 1: b(N=50000) = 1.2126 ? PASS

Test 1 [PASS] bN_convergence ? Test 1 [HARDCORE pass 2]: 8 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 2 [PASS] bN_monotonicity ? Test 2: Monotonicity violations: 0 / 499 ? PASS

Test 2 [PASS] bN_monotonicity ? Test 2 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Verdicts ? 79 tests: 66 PASS, 0 WARN, 13 FAIL, 0 DONE

Test 3 [PASS] bN_rate ? Test 3: ? = 0.1685 (empirical; no expected value), R?=0.9895 ? PASS

Test 3 [PASS] bN_rate ? Test 3 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 4 [FAIL] gue_ks_full ? Test 4: KS D=0.0216, p=0.0 ? WARN

Test 4 [PASS] gue_ks_full ? Test 4 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 5 [FAIL] gue_ks_highT ? Test 5: Best high-T p-value=0.0 ? WARN

Test 5 [PASS] gue_ks_highT ? Test 5 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 6 [FAIL] chi2_hist ? Test 6: ?? p=0.0 ? WARN

Test 6 [PASS] chi2_hist ? Test 6 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 7 [PASS] decay_slope ? Test 7: Slope=-0.1504 (empirical), 95%CI=[-0.1594,-0.1414] ? PASS

Test 7 [PASS] decay_slope ? Test 7 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 8 [PASS] residuals ? Test 8: Runs=3 (exp=5.8) ? PASS

Test 8 [PASS] residuals ? Test 8 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 9 [PASS] bootstrap_ci ? Test 9: Bootstrap slope=-0.1746 (empirical), 95%CI=[-0.1895,-0.1552] ? PASS

Test 9 [PASS] bootstrap_ci ? Test 9 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 10 [PASS] cross_validation ? Test 10: Max deviation=8.5% ? PASS

Test 10 [PASS] cross_validation ? Test 10 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 11 [FAIL] anderson_darling ? Test 11: A?=61.4924, MC p=0.0000 ? WARN (best high-T subrange T>31351 also rejects, p=0.0000; see height-resolved table)

Test 11 [PASS] anderson_darling ? Test 11 [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

Test 12 [FAIL] two_sample_ks ? Test 12: 2-sample KS D=0.0265, p=0.005252 ? WARN

Test 12 [FAIL] two_sample_ks ? Test 12 [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

Test 13 [PASS] number_variance ? Test 13: data closer to GUE in 9/9 L-values ? PASS

Test 13 [PASS] number_variance ? Test 13 [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Test 14 [PASS] spectral_rigidity ? Test 14: data closer to GUE in 9/9 L-values ? PASS

Test 14 [PASS] spectral_rigidity ? Test 14 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 15 [PASS] ab_construction ? Test 15: Hermiticity=OK, ?_TRB=0.1037, flux_v1=-0.0 [exp -0.0], flux_empty=0.0 ? PASS

Test 15 [PASS] ab_construction ? Test 15b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 16 [PASS] ab_gue_class ? Test 16 [MONTGOMERY] FINAL (last ?=0.5): ?r?=0.594 ? PASS

Test 16 [PASS] ab_gue_class ? Test 16b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 17 [PASS] connes_self_duality ? Test 17: zero_modes(=?=1/2)=4 (expected 4) ? PASS

Test 17 [PASS] connes_self_duality ? Test 17b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 18 [PASS] chiral_AIII ? Test 18: chiral defect (=?=1/2)=0.0 [exp 0.000000], (=?=1/3)=0.0058 [exp ?0] ? PASS

Test 18 [PASS] chiral_AIII ? Test 18b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Test 19 [PASS] dirac_cone ? Test 19: 1/L scaling R?=0.9997, v_F(2?)=1.8998, v_F(?)=3.7995 ? PASS

Test 19 [PASS] dirac_cone ? Test 19b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 19 [PASS] dirac_cone ? Test 19c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS

Test 19 [PASS] dirac_cone ? Test 19d [AUDIT pass 4]: 8 sub-checks, 0 failed ? PASS

Test 20 [PASS] chern_tknn ? Test 20: C? anchors {1/4,1/3,1/5}=1,1,1 [exp 1,1,1], ?=1/2 min gap=0.0 (Dirac touching), mass branch C=0/0 ? PASS

Test 20 [PASS] chern_tknn ? Test 20b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 21 [PASS] gamma_phase ? Test 21: arg(?*)=89.874° (?90°, deviation=0.126°) ? PASS

Test 21 [PASS] gamma_phase ? Test 21b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 22 [PASS] ab_phase ? Test 22: ?_AB=0.4487989505 = ?/7 (exact) ? PASS

Test 22 [PASS] ab_phase ? Test 22b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 23 [PASS] fractal_factor ? Test 23: |CF_formula|=1.0 (=1 by construction), c_AB=0.02062 (?0.02063) ? PASS

Test 23 [PASS] fractal_factor ? Test 23b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 24 [PASS] dirac_string_flux ? Test 24: vortex flux OK=1/1, empty OK=5040/5040, max_empty=0.0 ? PASS

Test 24 [PASS] dirac_string_flux ? Test 24b [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

Test 25 [PASS] byers_yang ? Test 25: ?(q=1?0)=0.0 [exp ~0], ?(q=0.3?0)=0.0032 [exp >1e-3] ? PASS

Test 25 [PASS] byers_yang ? Test 25b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 26 [PASS] pbc_torus ? Test 26: flux_v=-0.0 [exp -0.0], empty_ok=5040/5183, herm=OK, ?r?=0.6009 ? PASS

Test 26 [PASS] pbc_torus ? Test 26b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Test 27 [FAIL] binary_chiral ? EXCEPTION: UndefinedVarError: 'R_POISSON' not defined in 'Main'

Suggestion: check for spelling errors or missing imports.

Test 27 [PASS] binary_chiral ? Test 27b [HARDCORE pass 2]: 8 sub-checks, 0 failed ? PASS

Test 28 [FAIL] f_gue_merit ? Test 28: f_GUE=0.881, ??_data=0.6115 ? WARN

Test 28 [FAIL] f_gue_merit ? Test 28b [HARDCORE pass 2]: 2 sub-checks, 1 failed ? WARN

Test 29 [PASS] dirac_dip ? Test 29: ?(?=0.5)=0.0193 vs ?(0.403)=0.1944 / ?(0.597)=0.1944 ? PASS

Test 29 [FAIL] dirac_dip ? Test 29b [HARDCORE pass 2]: 3 sub-checks, 1 failed ? WARN

Test 29 [PASS] dirac_dip ? Test 29c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS

Test 30 [PASS] vf_scaling ? Test 30: v_F=1.798, R?=0.9977 ? PASS

Test 30 [PASS] vf_scaling ? Test 30b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 31 [PASS] hatano_nelson ? Test 31: max|Im(E)|=0.6397, ?r?_NH=0.8938 ? PASS

Test 31 [PASS] hatano_nelson ? Test 31b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 31 [PASS] hatano_nelson ? Test 31c [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS

Test 32 [PASS] rmean_bootstrap ? Test 32: ?r?=0.5991-0.0075 (q=1.0, n=5, W=1.0) vs GUE 0.5992 ? PASS

Test 32 [PASS] rmean_bootstrap ? Test 32b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 33 [PASS] l_scaling_rmean ? Test 33: L-scaling to L=80 0.434,0.556,0.593,0.6,0.6 ? PASS

Test 33 [PASS] l_scaling_rmean ? Test 33b (hardcore): plateau weighted ?r?=0.6004, ??/dof=0.01 ? PASS

Test 34 [PASS] direct_vs_zeta ? Test 34: KS p=0.0, ?|?R?|?=0.0374, d_GUE=0.796 ? PASS

Test 34 [FAIL] direct_vs_zeta ? Test 34b [HARDCORE pass 2]: 4 sub-checks, 1 failed ? WARN

Test 35 [FAIL] form_factor_Kt ? Test 35: K(t) ramp+plateau ? RMS_GUE=0.4193, corr_GUE=0.8889 ? WARN

Test 35 [FAIL] form_factor_Kt ? Test 35b [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

Test 36 [PASS] byte_robust ? Test 36: byte robust ? r_256=0.5894, |?r|_multi=0.001 (120 windows), |?r|_1win=0.012 (diag), H=7.9991 bits, ??_z=-0.234, n=27997 ? PASS

Test 36 [PASS] byte_robust ? Test 36: byte robust ? r_256=0.5756, |?r|_multi=0.0009 (120 windows), |?r|_1win=0.0061 (diag), H=7.9991 bits, ??_z=-0.234, n=55288 ? PASS

Test 37 [PASS] half_factorial_gamma ? Test 37: (1/2)! = ??/2 [rel.err 0.0], 32/?? [rel.err 0.0], ?p?=1 [|?=0.0], uniqueness ? ? PASS

Test 37 [PASS] half_factorial_gamma ? Test 37b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 38 [PASS] curie_point ? Test 38 (v23 three-pass) t38: PASS • Curie 3-pass (72?+96?+38c) ? 486.9 min, 346 spectra in cache

Execution steps (computation-log section markers)

[23:46:49.053] [+ 0.081s] ? run_all_tests START

[23:46:49.141] [+ 0.170s] ? Test 1 ? pass 1 (primary: standard implementation)

[23:46:50.101] [+ 1.129s] pass 1 result: PASS (elapsed: 0.96s)

[23:46:50.101] [+ 1.129s] ? Test 1 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:47:03.102] [+ 14.130s] pass 2 result: PASS (elapsed: 13.00s)

[23:47:03.102] [+ 14.130s] Test 1 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:47:03.751] [+ 14.779s] ? Test 2 ? pass 1 (primary: standard implementation)

[23:47:14.727] [+ 25.755s] pass 1 result: PASS (elapsed: 10.98s)

[23:47:14.727] [+ 25.755s] ? Test 2 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:47:20.626] [+ 31.654s] pass 2 result: PASS (elapsed: 5.90s)

[23:47:20.626] [+ 31.654s] Test 2 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:47:21.277] [+ 32.305s] ? Test 3 ? pass 1 (primary: standard implementation)

[23:47:21.598] [+ 32.627s] pass 1 result: PASS (elapsed: 0.32s)

[23:47:21.598] [+ 32.627s] ? Test 3 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:47:27.415] [+ 38.443s] pass 2 result: PASS (elapsed: 5.82s)

[23:47:27.415] [+ 38.444s] Test 3 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:47:28.065] [+ 39.093s] ? Test 4 ? pass 1 (primary: standard implementation)

[23:52:57.704] [+368.732s] pass 1 result: FAIL/WARN (elapsed: 329.64s)

[23:52:57.704] [+368.733s] ? Test 4 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:53:06.749] [+377.777s] pass 2 result: PASS (elapsed: 9.04s)

[23:53:06.749] [+377.778s] Test 4 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[23:53:08.419] [+379.447s] ? Test 5 ? pass 1 (primary: standard implementation)

[23:53:09.153] [+380.181s] pass 1 result: FAIL/WARN (elapsed: 0.73s)

[23:53:09.153] [+380.182s] ? Test 5 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:53:14.150] [+385.178s] pass 2 result: PASS (elapsed: 5.00s)

[23:53:14.150] [+385.178s] Test 5 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[23:53:14.998] [+386.026s] ? Test 6 ? pass 1 (primary: standard implementation)

[23:53:15.580] [+386.609s] pass 1 result: FAIL/WARN (elapsed: 0.58s)

[23:53:15.581] [+386.609s] ? Test 6 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:53:20.392] [+391.421s] pass 2 result: PASS (elapsed: 4.81s)

[23:53:20.392] [+391.421s] Test 6 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[23:53:21.221] [+392.249s] ? Test 7 ? pass 1 (primary: standard implementation)

[23:53:21.563] [+392.591s] pass 1 result: PASS (elapsed: 0.34s)

[23:53:21.563] [+392.591s] ? Test 7 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:53:28.609] [+399.637s] pass 2 result: PASS (elapsed: 7.05s)

[23:53:28.609] [+399.637s] Test 7 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:53:29.433] [+400.462s] ? Test 8 ? pass 1 (primary: standard implementation)

[23:53:29.703] [+400.731s] pass 1 result: PASS (elapsed: 0.27s)

[23:53:29.703] [+400.731s] ? Test 8 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:53:36.917] [+407.945s] pass 2 result: PASS (elapsed: 7.21s)

[23:53:36.917] [+407.945s] Test 8 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:53:37.821] [+408.849s] ? Test 9 ? pass 1 (primary: standard implementation)

[23:53:38.031] [+409.059s] pass 1 result: PASS (elapsed: 0.21s)

[23:53:38.031] [+409.059s] ? Test 9 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:53:44.802] [+415.831s] pass 2 result: PASS (elapsed: 6.77s)

[23:53:44.803] [+415.831s] Test 9 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:53:45.593] [+416.622s] ? Test 10 ? pass 1 (primary: standard implementation)

[23:53:46.068] [+417.096s] pass 1 result: PASS (elapsed: 0.47s)

[23:53:46.068] [+417.096s] ? Test 10 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:53:53.027] [+424.056s] pass 2 result: PASS (elapsed: 6.96s)

[23:53:53.027] [+424.056s] Test 10 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:53:53.811] [+424.839s] ? Test 11 ? pass 1 (primary: standard implementation)

[23:55:18.654] [+509.682s] pass 1 result: FAIL/WARN (elapsed: 84.84s)

[23:55:18.654] [+509.682s] ? Test 11 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:56:58.575] [+609.604s] pass 2 result: PASS (elapsed: 99.92s)

[23:56:58.575] [+609.604s] Test 11 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[23:56:59.260] [+610.289s] ? Test 12 ? pass 1 (primary: standard implementation)

[23:56:59.394] [+610.422s] pass 1 result: FAIL/WARN (elapsed: 0.13s)

[23:56:59.394] [+610.422s] ? Test 12 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:57:03.401] [+614.429s] pass 2 result: FAIL/WARN (elapsed: 4.01s)

[23:57:03.401] [+614.429s] Test 12 two-pass summary: pass1=FAIL pass2=FAIL (hardcore) ? OVERALL FAIL

[23:57:04.082] [+615.110s] ? Test 13 ? pass 1 (primary: standard implementation)

[23:57:04.357] [+615.385s] pass 1 result: PASS (elapsed: 0.27s)

[23:57:04.357] [+615.385s] ? Test 13 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:57:10.480] [+621.508s] pass 2 result: PASS (elapsed: 6.12s)

[23:57:10.480] [+621.508s] Test 13 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:57:11.199] [+622.227s] ? Test 14 ? pass 1 (primary: standard implementation)

[23:57:11.705] [+622.733s] pass 1 result: PASS (elapsed: 0.51s)

[23:57:11.705] [+622.733s] ? Test 14 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[23:57:18.173] [+629.201s] pass 2 result: PASS (elapsed: 6.47s)

[23:57:18.173] [+629.201s] Test 14 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[23:57:18.882] [+629.910s] ? Test 15 ? pass 1 (primary: 72×72, Nv=[4])

[23:57:22.443] [+633.472s] pass 1 result: PASS (elapsed: 3.56s)

[23:57:22.444] [+633.472s] ? Test 15 ? pass 2 (secondary: 96×96, Nv=[2])

[23:57:36.325] [+647.353s] pass 2 result: PASS (elapsed: 13.88s)

[23:57:36.325] [+647.353s] Test 15 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[23:57:37.019] [+648.047s] ? Test 16 ? pass 1 (primary: 72×72, Nv=[4])

[23:57:37.289] [+648.317s] ? Test 16 single-?: ?=0.500000 (?=0.5000 (default))

[00:00:52.674] [+843.702s] pass 1 result: PASS (elapsed: 195.66s)

[00:00:52.674] [+843.702s] ? Test 16 ? pass 2 (secondary: 96×96, Nv=[2])

[00:25:09.091] [+2300.119s] pass 2 result: PASS (elapsed: 1456.42s)

[00:25:09.091] [+2300.119s] Test 16 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[00:25:10.161] [+2301.189s] ? Test 17 ? pass 1 (primary: 72×72, Nv=[4])

[00:32:11.010] [+2722.038s] pass 1 result: PASS (elapsed: 420.85s)

[00:32:11.010] [+2722.039s] ? Test 17 ? pass 2 (secondary: 96×96, Nv=[2])

[00:55:53.775] [+4144.804s] pass 2 result: PASS (elapsed: 1422.76s)

[00:55:53.777] [+4144.805s] Test 17 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[00:55:58.273] [+4149.301s] ? Test 18 ? pass 1 (primary: 72×72, Nv=[4])

[01:00:13.907] [+4404.936s] pass 1 result: PASS (elapsed: 255.63s)

[01:00:13.908] [+4404.936s] ? Test 18 ? pass 2 (secondary: 96×96, Nv=[2])

[01:01:22.413] [+4473.442s] pass 2 result: PASS (elapsed: 68.51s)

[01:01:22.413] [+4473.442s] Test 18 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[01:01:23.562] [+4474.590s] ? Test 19 ? pass 1 (primary: 72×72, Nv=[4])

[01:03:00.683] [+4571.711s] pass 1 result: PASS (elapsed: 97.12s)

[01:03:00.683] [+4571.711s] ? Test 19 ? pass 2 (secondary: 96×96, Nv=[2])

[01:16:29.739] [+5380.767s] pass 2 result: PASS (elapsed: 809.06s)

[01:16:29.739] [+5380.768s] ? Test 19 ? pass 3 (tertiary: SERIES ? real report series, 96×96 regime)

[01:23:43.291] [+5814.319s] ? Test 19 ? pass 4 (quaternary: AUDIT ? twist spectroscopy + eigenvector index, 96×96 regime)

[01:25:00.806] [+5891.834s] ? Test 20 ? pass 1 (primary: 72✕72, Nv=[4])

[01:25:01.663] [+5892.692s] pass 1 result: PASS (elapsed: 0.86s)

[01:25:01.664] [+5892.692s] ? Test 20 ? pass 2 (secondary: 96✕06, Nv=[2])

[01:25:29.607] [+5920.635s] pass 2 result: PASS (elapsed: 27.94s)

[01:25:29.607] [+5920.635s] Test 20 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[01:25:30.660] [+5921.689s] ? Test 21 ? pass 1 (primary: 72✕72, Nv=[4])

[01:25:31.071] [+5922.099s] pass 1 result: PASS (elapsed: 0.41s)

[01:25:31.071] [+5922.099s] ? Test 21 ? pass 2 (secondary: 96✕06, Nv=[2])

[01:25:40.268] [+5931.297s] pass 2 result: PASS (elapsed: 9.20s)

[01:25:40.268] [+5931.297s] Test 21 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[01:25:41.304] [+5932.332s] ? Test 22 ? pass 1 (primary: 72✕72, Nv=[4])

[01:25:41.570] [+5932.599s] pass 1 result: PASS (elapsed: 0.27s)

[01:25:41.571] [+5932.599s] ? Test 22 ? pass 2 (secondary: 96✕06, Nv=[2])

[01:25:50.745] [+5941.774s] pass 2 result: PASS (elapsed: 9.17s)

[01:25:50.745] [+5941.774s] Test 22 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[01:25:51.801] [+5942.829s] ? Test 23 ? pass 1 (primary: 72✕72, Nv=[4])

[01:25:52.306] [+5943.334s] pass 1 result: PASS (elapsed: 0.51s)

[01:25:52.306] [+5943.334s] ? Test 23 ? pass 2 (secondary: 96✕06, Nv=[2])

[01:26:01.167] [+5952.195s] pass 2 result: PASS (elapsed: 8.86s)

[01:26:01.167] [+5952.195s] Test 23 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[01:26:02.205] [+5953.233s] ? Test 24 ? pass 1 (primary: 72✕72, Nv=[4])

[01:26:04.492] [+5955.520s] pass 1 result: PASS (elapsed: 2.29s)

[01:26:04.492] [+5955.520s] ? Test 24 ? pass 2 (secondary: 96✕06, Nv=[2])

[01:26:53.431] [+6004.459s] pass 2 result: PASS (elapsed: 48.94s)

[01:26:53.431] [+6004.459s] Test 24 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[01:26:54.375] [+6005.403s] ? Test 25 ? pass 1 (primary: 72✕72, Nv=[4])

[01:29:45.658] [+6176.686s] pass 1 result: PASS (elapsed: 171.28s)

[01:29:45.658] [+6176.687s] ? Test 25 ? pass 2 (secondary: 96✕06, Nv=[2])

[02:34:42.641] [+10073.669s] pass 2 result: PASS (elapsed: 3896.98s)

[02:34:42.641] [+10073.669s] Test 25 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[02:34:43.847] [+10074.876s] ? Test 26 ? pass 1 (primary: 72✕72, Nv=[4])

[02:35:51.050] [+10142.078s] pass 1 result: PASS (elapsed: 67.20s)

[02:35:51.050] [+10142.078s] ? Test 26 ? pass 2 (secondary: 96✕06, Nv=[2])

[03:07:11.089] [+12022.117s] pass 2 result: PASS (elapsed: 1880.04s)

[03:07:11.089] [+12022.117s] Test 26 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[03:07:12.359] [+12023.387s] ? Test 27 ? pass 1 (primary: 72✖72, Nv=[4])

[03:07:30.520] [+12041.549s] pass 1 result: FAIL/WARN (elapsed: 18.16s)

[03:07:30.520] [+12041.549s] ? Test 27 ? pass 2 (secondary: 96✖06, Nv=[2])

[03:10:06.429] [+12197.457s] pass 2 result: PASS (elapsed: 155.91s)

[03:10:06.429] [+12197.457s] Test 27 two-pass summary: pass1=FAIL pass2=PASS ? OVERALL FAIL

[03:10:07.363] [+12198.391s] ? Test 28 ? pass 1 (primary: 72✖72, Nv=[4])

[03:10:53.830] [+12244.858s] pass 1 result: FAIL/WARN (elapsed: 46.47s)

[03:10:53.830] [+12244.859s] ? Test 28 ? pass 2 (secondary: 96✖06, Nv=[2])

[03:33:32.361] [+13603.389s] pass 2 result: FAIL/WARN (elapsed: 1358.53s)

[03:33:32.361] [+13603.389s] Test 28 two-pass summary: pass1=FAIL pass2=FAIL ? OVERALL FAIL

[03:33:33.339] [+13604.367s] ? Test 29 ? pass 1 (primary: 72✖72, Nv=[4])

[03:35:42.492] [+13733.520s] pass 1 result: PASS (elapsed: 129.15s)

[03:35:42.492] [+13733.520s] ? Test 29 ? pass 2 (secondary: 96✖06, Nv=[2])

[03:49:18.967] [+14549.996s] pass 2 result: FAIL/WARN (elapsed: 816.48s)

[03:49:18.968] [+14549.996s] ? Test 29 ? pass 3 (tertiary: SERIES ? real report series, 96✖06 regime)

[04:28:43.626] [+16914.654s] ? Test 30 ? pass 1 (primary: 72✖72, Nv=[4])

[04:30:03.314] [+16994.342s] pass 1 result: PASS (elapsed: 79.69s)

[04:30:03.314] [+16994.342s] ? Test 30 ? pass 2 (secondary: 96✖06, Nv=[2])

[04:40:31.149] [+17622.177s] pass 2 result: PASS (elapsed: 627.83s)

[04:40:31.149] [+17622.178s] Test 30 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[04:40:32.099] [+17623.127s] ? Test 31 ? pass 1 (primary: 72✖72, Nv=[4])

[04:43:47.935] [+17818.964s] pass 1 result: PASS (elapsed: 195.84s)

[04:43:47.936] [+17818.964s] ? Test 31 ? pass 2 (secondary: 96✖06, Nv=[2])

[05:25:20.799] [+20311.827s] pass 2 result: PASS (elapsed: 2492.86s)

[05:25:20.799] [+20311.827s] ? Test 31 ? pass 3 (tertiary: SERIES ? real report series, 96✖06 regime)

[05:30:07.638] [+20598.666s] ? Test 32 ? pass 1 (primary: 72✖72, Nv=[4])

[05:37:37.908] [+21048.937s] pass 1 result: PASS (elapsed: 450.27s)

[05:37:37.909] [+21048.937s] ? Test 32 ? pass 2 (secondary: 96✖06, Nv=[2])

[06:47:24.642] [+25235.670s] pass 2 result: PASS (elapsed: 4186.73s)

[06:47:24.642] [+25235.671s] Test 32 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[06:47:25.815] [+25236.844s] ? Test 33 ? pass 1 (primary: 72✖72, Nv=[4])

[07:09:25.403] [+26556.431s] pass 1 result: PASS (elapsed: 1319.59s)

[07:09:25.404] [+26556.432s] ? Test 33 ? pass 2 (secondary: 96✖06, Nv=[2])

[07:26:40.230] [+27591.258s] pass 2 result: PASS (elapsed: 1034.83s)

[07:26:40.230] [+27591.258s] Test 33 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[07:26:41.241] [+27592.269s] ? Test 34 ? pass 1 (primary: 72✕72, Nv=[4])

[07:30:23.296] [+27814.324s] pass 1 result: PASS (elapsed: 222.06s)

[07:30:23.296] [+27814.324s] ? Test 34 ? pass 2 (secondary: 96✕96, Nv=[2])

[07:52:07.208] [+29118.237s] pass 2 result: FAIL/WARN (elapsed: 1303.91s)

[07:52:07.209] [+29118.237s] Test 34 two-pass summary: pass1=PASS pass2=FAIL ? OVERALL FAIL

[07:52:08.147] [+29119.176s] ? Test 35 ? pass 1 (primary: 72✕72, Nv=[4])

[07:55:51.681] [+29342.709s] pass 1 result: FAIL/WARN (elapsed: 223.53s)

[07:55:51.681] [+29342.709s] ? Test 35 ? pass 2 (secondary: 96✕96, Nv=[2])

[08:18:03.989] [+30675.017s] pass 2 result: FAIL/WARN (elapsed: 1332.31s)

[08:18:03.989] [+30675.017s] Test 35 two-pass summary: pass1=FAIL pass2=FAIL ? OVERALL FAIL

[08:18:04.978] [+30676.007s] ? Test 36 ? pass 1 (primary: 72✕72, Nv=[4])

[08:25:15.606] [+31106.634s] pass 1 result: PASS (elapsed: 430.63s)

[08:25:15.607] [+31106.636s] ? Test 36 ? pass 2 (secondary: 96✕96, Nv=[2])

[09:08:55.777] [+33726.806s] pass 2 result: PASS (elapsed: 2620.17s)

[09:08:55.778] [+33726.806s] Test 36 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[09:08:56.803] [+33727.831s] ? Test 37 ? pass 1 (primary: 72✕72, Nv=[4])

[09:09:07.239] [+33738.267s] pass 1 result: PASS (elapsed: 10.44s)

[09:09:07.239] [+33738.267s] ? Test 37 ? pass 2 (secondary: 96✕96, Nv=[2])

[09:09:14.797] [+33745.826s] pass 2 result: PASS (elapsed: 7.56s)

[09:09:14.797] [+33745.826s] Test 37 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[09:09:15.732] [+33746.760s] ? Test 38 ? pass 1 (primary: 72✕72, Nv=[4])

[17:16:11.648] [+61287.429s] pass 1 result: PASS (elapsed: 27540.67s)

test_01_bN_convergence [PASS] reports/ + logs/ copied; plots live in ../test_01_bN_convergence/plots/

test_01_bN_convergence [PASS] reports/ + logs/ copied; plots live in ../test_01_bN_convergence/plots/

test_02_bN_monotonicity [PASS] reports/ + logs/ copied; plots live in ../test_02_bN_monotonicity/plots/

test_02_bN_monotonicity [PASS] reports/ + logs/ copied; plots live in ../test_02_bN_monotonicity/plots/

Artifact map

test_03_bN_rate [PASS] reports/ + logs/ copied; plots live in ../test_03_bN_rate/plots/

test_03_bN_rate [PASS] reports/ + logs/ copied; plots live in ../test_03_bN_rate/plots/

test_04_gue_ks_full [FAIL] reports/ + logs/ copied; plots live in ../test_04_gue_ks_full/plots/

test_04_gue_ks_full [PASS] reports/ + logs/ copied; plots live in ../test_04_gue_ks_full/plots/

test_05_gue_ks_highT [FAIL] reports/ + logs/ copied; plots live in ../test_05_gue_ks_highT/plots/

test_05_gue_ks_highT [PASS] reports/ + logs/ copied; plots live in ../test_05_gue_ks_highT/plots/

test_06_chi2_hist [FAIL] reports/ + logs/ copied; plots live in ../test_06_chi2_hist/plots/

test_06_chi2_hist [PASS] reports/ + logs/ copied; plots live in ../test_06_chi2_hist/plots/

test_07_decay_slope [PASS] reports/ + logs/ copied; plots live in ../test_07_decay_slope/plots/

test_07_decay_slope [PASS] reports/ + logs/ copied; plots live in ../test_07_decay_slope/plots/

test_08_residuals [PASS] reports/ + logs/ copied; plots live in ../test_08_residuals/plots/

test_08_residuals [PASS] reports/ + logs/ copied; plots live in ../test_08_residuals/plots/

test_09_bootstrap_ci [PASS] reports/ + logs/ copied; plots live in ../test_09_bootstrap_ci/plots/

test_09_bootstrap_ci [PASS] reports/ + logs/ copied; plots live in ../test_09_bootstrap_ci/plots/

test_10_cross_validation [PASS] reports/ + logs/ copied; plots live in ../test_10_cross_validation/plots/

test_10_cross_validation [PASS] reports/ + logs/ copied; plots live in ../test_10_cross_validation/plots/

test_11_anderson_darling [FAIL] reports/ + logs/ copied; plots live in ../test_11_anderson_darling/plots/

test_11_anderson_darling [PASS] reports/ + logs/ copied; plots live in ../test_11_anderson_darling/plots/

test_12_two_sample_ks [FAIL] reports/ + logs/ copied; plots live in ../test_12_two_sample_ks/plots/

test_12_two_sample_ks [FAIL] reports/ + logs/ copied; plots live in ../test_12_two_sample_ks/plots/

test_13_number_variance [PASS] reports/ + logs/ copied; plots live in ../test_13_number_variance/plots/

test_13_number_variance [PASS] reports/ + logs/ copied; plots live in ../test_13_number_variance/plots/

test_14_spectral_rigidity [PASS] reports/ + logs/ copied; plots live in ../test_14_spectral_rigidity/plots/

test_14_spectral_rigidity [PASS] reports/ + logs/ copied; plots live in ../test_14_spectral_rigidity/plots/

test_15_ab_construction [PASS] reports/ + logs/ copied; plots live in ../test_15_ab_construction/plots/

test_15_ab_construction [PASS] reports/ + logs/ copied; plots live in ../test_15_ab_construction/plots/

test_16_ab_gue_class [PASS] reports/ + logs/ copied; plots live in ../test_16_ab_gue_class/plots/

test_16_ab_gue_class [PASS] reports/ + logs/ copied; plots live in ../test_16_ab_gue_class/plots/

test_17_connes_self_duality [PASS] reports/ + logs/ copied; plots live in ../test_17_connes_self_duality/plots/

test_17_connes_self_duality [PASS] reports/ + logs/ copied; plots live in ../test_17_connes_self_duality/plots/

test_18_chiral_AIII [PASS] reports/ + logs/ copied; plots live in ../test_18_chiral_AIII/plots/

test_18_chiral_AIII [PASS] reports/ + logs/ copied; plots live in ../test_18_chiral_AIII/plots/

test_19_dirac_cone [PASS] reports/ + logs/ copied; plots live in ../test_19_dirac_cone/plots/

test_19_dirac_cone [PASS] reports/ + logs/ copied; plots live in ../test_19_dirac_cone/plots/

test_19_dirac_cone [PASS] reports/ + logs/ copied; plots live in ../test_19_dirac_cone/plots/

test_19_dirac_cone [PASS] reports/ + logs/ copied; plots live in ../test_19_dirac_cone/plots/

test_20_chern_tknn [PASS] reports/ + logs/ copied; plots live in ../test_20_chern_tknn/plots/

test_20_chern_tknn [PASS] reports/ + logs/ copied; plots live in ../test_20_chern_tknn/plots/

test_21_gamma_phase [PASS] reports/ + logs/ copied; plots live in ../test_21_gamma_phase/plots/

test_21_gamma_phase [PASS] reports/ + logs/ copied; plots live in ../test_21_gamma_phase/plots/

test_22_ab_phase [PASS] reports/ + logs/ copied; plots live in ../test_22_ab_phase/plots/

test_22_ab_phase [PASS] reports/ + logs/ copied; plots live in ../test_22_ab_phase/plots/

test_23_fractal_factor [PASS] reports/ + logs/ copied; plots live in ../test_23_fractal_factor/plots/

test_23_fractal_factor [PASS] reports/ + logs/ copied; plots live in ../test_23_fractal_factor/plots/

test_24_dirac_string_flux [PASS] reports/ + logs/ copied; plots live in ../test_24_dirac_string_flux/plots/

test_24_dirac_string_flux [PASS] reports/ + logs/ copied; plots live in ../test_24_dirac_string_flux/plots/

test_25_byers_yang [PASS] reports/ + logs/ copied; plots live in ../test_25_byers_yang/plots/

test_25_byers_yang [PASS] reports/ + logs/ copied; plots live in ../test_25_byers_yang/plots/

test_26_pbc_torus [PASS] reports/ + logs/ copied; plots live in ../test_26_pbc_torus/plots/

test_26_pbc_torus [PASS] reports/ + logs/ copied; plots live in ../test_26_pbc_torus/plots/

test_27_binary_chiral [FAIL] reports/ + logs/ copied; plots live in ../test_27_binary_chiral/plots/

test_27_binary_chiral [PASS] reports/ + logs/ copied; plots live in ../test_27_binary_chiral/plots/

test_28_f_gue_merit [FAIL] reports/ + logs/ copied; plots live in ../test_28_f_gue_merit/plots/

test_28_f_gue_merit [FAIL] reports/ + logs/ copied; plots live in ../test_28_f_gue_merit/plots/

test_29_dirac_dip [PASS] reports/ + logs/ copied; plots live in ../test_29_dirac_dip/plots/

test_29_dirac_dip [FAIL] reports/ + logs/ copied; plots live in ../test_29_dirac_dip/plots/

test_29_dirac_dip [PASS] reports/ + logs/ copied; plots live in ../test_29_dirac_dip/plots/

test_30_vf_scaling [PASS] reports/ + logs/ copied; plots live in ../test_30_vf_scaling/plots/

test_30_vf_scaling [PASS] reports/ + logs/ copied; plots live in ../test_30_vf_scaling/plots/

test_31_hatano_nelson [PASS] reports/ + logs/ copied; plots live in ../test_31_hatano_nelson/plots/

test_31_hatano_nelson [PASS] reports/ + logs/ copied; plots live in ../test_31_hatano_nelson/plots/

test_31_hatano_nelson [PASS] reports/ + logs/ copied; plots live in ../test_31_hatano_nelson/plots/

test_32_rmean_bootstrap [PASS] reports/ + logs/ copied; plots live in ../test_32_rmean_bootstrap/plots/

test_32_rmean_bootstrap [PASS] reports/ + logs/ copied; plots live in ../test_32_rmean_bootstrap/plots/

test_33_l_scaling_rmean [PASS] reports/ + logs/ copied; plots live in ../test_33_l_scaling_rmean/plots/

test_33_l_scaling_rmean [PASS] reports/ + logs/ copied; plots live in ../test_33_l_scaling_rmean/plots/

test_34_direct_vs_zeta [PASS] reports/ + logs/ copied; plots live in ../test_34_direct_vs_zeta/plots/

test_34_direct_vs_zeta [FAIL] reports/ + logs/ copied; plots live in ../test_34_direct_vs_zeta/plots/

test_35_form_factor_Kt [FAIL] reports/ + logs/ copied; plots live in ../test_35_form_factor_Kt/plots/

test_35_form_factor_Kt [FAIL] reports/ + logs/ copied; plots live in ../test_35_form_factor_Kt/plots/

test_36_byte_robust [PASS] reports/ + logs/ copied; plots live in ../test_36_byte_robust/plots/

test_36_byte_robust [PASS] reports/ + logs/ copied; plots live in ../test_36_byte_robust/plots/

test_37_half_factorial_gamma [PASS] reports/ + logs/ copied; plots live in ../test_37_half_factorial_gamma/plots/

test_37_half_factorial_gamma [PASS] reports/ + logs/ copied; plots live in ../test_37_half_factorial_gamma/plots/

test_38_curie_point [PASS] reports/ + logs/ copied; plots live in ../test_38_curie_point/plots/

Log files: logs/full_run_log.txt (THE consolidated log: every test's computation log + console capture in one file), logs/computation_log.txt / computation_log_full.txt (every computation log)

Run index: ../index.html ? verdict table with direct links to every per-test report.

Full per-test reports (gathered inline ? one-file read of the whole run)

Every test's full report (method, verdict-table content, plot list, verification guide) is embedded below in run order. Computation logs and console captures are consolidated separately

Test 1 ? bN_convergence [PASS]

Verdict: PASS | Generated: 2026-09-14 23:46:52 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

b(N) = (1/N)?|?_k???_k| measures convergence of Gram-point approximations to true zeta zeros. EXACT-formula family: tolerance scale = floating-point round-off.

What this test verifies

METHOD: b(N) = (1/N)?|?_k???_k| measures convergence of Gram-point approximations to true zeta zeros. EXACT-formula family: tolerance scale = floating-point round-off.

Test 1 [HARDCORE pass 2]: 8 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)

- 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 1 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide for this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:46:52 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

b(N) = (1/N)?|?_k???_k| measures convergence of Gram-point approximations to true zeta zeros. EXACT-formula family: tolerance scale = floating-point round-off.

Test 1 ? bN_convergence [PASS]

METHOD: b(N) = (1/N)?|?_k???_k| measures convergence of Gram-point approximations to true zeta zeros. EXACT-formula family: tolerance scale = floating-point round-off.

Test 1 [HARDCORE pass 2]: 8 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

What this test verifies

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)

- 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)

Result

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 1 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Test 2 ? bN_monotonicity [PASS]

Verdict: PASS | Generated: 2026-09-14 23:47:15 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Checks b(N) does not increase along the refinement sequence ? an algebraic consistency property (binary verdict).

METHOD: Checks b(N) does not increase along the refinement sequence ? an algebraic consistency property (binary verdict).

Test 2 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)

Plots/animation.gif ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 2 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

Independent verification guide

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:47:15 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 2 ? bN_monotonicity [PASS]

Checks b(N) does not increase along the refinement sequence ? an algebraic consistency property (binary verdict).

METHOD: Checks b(N) does not increase along the refinement sequence ? an algebraic consistency property (binary verdict).

What this test verifies

Test 2 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)

Plots/animation.gif ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 2 --no-two-pass`

Plots

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Independent verification guide

Test 3 ? bN_rate [PASS]

Verdict: PASS | Generated: 2026-09-14 23:47:21 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

log-log regression of b(N): the slope is a formula-derived quantity with R? as the goodness yardstick.

METHOD: log-log regression of b(N): the slope is a formula-derived quantity with R? as the goodness yardstick.

Result

Test 3 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 3 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:47:21 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 3 ? bN_rate [PASS]

log-log regression of b(N): the slope is a formula-derived quantity with R? as the goodness yardstick.

METHOD: log-log regression of b(N): the slope is a formula-derived quantity with R? as the goodness yardstick.

What this test verifies

Test 3 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Result

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 3 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-14 23:52:57 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 4 ? gue_ks_full [FAIL]

What this test verifies

STATISTICAL family: one-sample KS of zero spacings vs GUE Wigner surmise. p-values depend on sample size and T-range ? finite-data effects are legal.

METHOD: STATISTICAL family: one-sample KS of zero spacings vs GUE Wigner surmise. p-values depend on sample size and T-range ? finite-data effects are legal.

Result

Test 4 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Plots

plots/plot_01.png ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 4 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:52:57 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 4 ? gue_ks_full [PASS]

STATISTICAL family: one-sample KS of zero spacings vs GUE Wigner surmise. p-values depend on sample size and T-range ? finite-data effects are legal.

METHOD: STATISTICAL family: one-sample KS of zero spacings vs GUE Wigner surmise. p-values depend on sample size and T-range ? finite-data effects are legal.

What this test verifies

Test 4 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Result

plots/plot_01.png ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 4 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-14 23:53:09 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Same, restricted to high T where GUE asymptotics hold better; still data-limited (embedded 50k zeros, T ? 40434 ? 10?).

METHOD: Same, restricted to high T where GUE asymptotics hold better; still data-limited (embedded 50k zeros, T ? 40434 ? 10?).

Test 5 ? gue_ks_highT [FAIL]

What this test verifies

Result

Test 5 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 5 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:53:09 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 5 ? gue_ks_highT [PASS]

Same, restricted to high T where GUE asymptotics hold better; still data-limited (embedded 50k zeros, T ? 40434 ? 10?).

What this test verifies to high T where GUE asymptotics hold better; still data-limited (embedded 50k zeros, T ? 40434 ? 10?).

Test 5 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 5 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-14 23:53:15 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

?? binned comparison of spacing histogram to Wigner-Dyson. Binning choices are part of the tolerance system.

Test 6 ? chi2_hist [FAIL]

Test 6 ?? binned comparison of spacing histogram to Wigner-Dyson. Binning choices are part of the tolerance system.

Test 6 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

What this test verifies

Result

Plots

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: julia ab_cloud_v23.jl --test 6 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:53:15 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 6 ? chi2_hist [PASS]

?? binned comparison of spacing histogram to Wigner-Dyson. Binning choices are part of the tolerance system.

What this test verifies

METHOD: ?? binned comparison of spacing histogram to Wigner-Dyson. Binning choices are part of the tolerance system.

Test 6 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 6 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-14 23:53:21 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Large-T decay of b(N): formula family (slope??0.5), verified via linear regression with R?.

Test 7 ? decay_slope [PASS]

METHOD: Large-T decay of b(N): formula family (slope??0.5), verified via linear regression with R?.

Test 7 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

What this test verifies

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Result

Plots

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 7 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Test 7 ? decay_slope [PASS]

Verdict: PASS | Generated: 2026-09-14 23:53:21 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies
What this test verifies: Large-T decay of b(N): formula family (slope??0.5), verified via linear regression with R?.

METHOD: Large-T decay of b(N): formula family (slope??0.5), verified via linear regression with R?.

Test 7 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)

Plots/animation.gif ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 7 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

Independent verification guide

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:53:29 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Residual structure of the log-log fit ? algebraic diagnostics.

Test 8 ? residuals [PASS]

METHOD: Residual structure of the log-log fit ? algebraic diagnostics.

What this test verifies

Test 8 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)

Plots/animation.gif ? animated reveal of every plot of this test

Result

Reproduce with: `julia ab_cloud_v23.jl --test 8 --no-two-pass`

Plots

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Independent verification guide

Test 8 ? residuals [PASS]

Verdict: PASS | Generated: 2026-09-14 23:53:29 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Residual structure of the log-log fit ? algebraic diagnostics.

METHOD: Residual structure of the log-log fit ? algebraic diagnostics.

Result

Test 8 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 8 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:53:38 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 9 ? bootstrap_ci [PASS]

Sampling-noise quantification: statistical system with CIs.

METHOD: Sampling-noise quantification: statistical system with CIs.

What this test verifies

Test 9 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Result

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 9 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-14 23:53:38 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 9 ? bootstrap_ci [PASS]

What this test verifies

Sampling-noise quantification: statistical system with CIs.

METHOD: Sampling-noise quantification: statistical system with CIs.

Result

Test 9 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Plots

plots/plot_01.png ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 9 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:53:46 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 10 ? cross_validation [PASS]

Statistical stability of the fit under data splits.

METHOD: Statistical stability of the fit under data splits.

What this test verifies

Test 10 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Result

plots/plot_01.png ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 10 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-14 23:53:46 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Statistical stability of the fit under data splits.

METHOD: Statistical stability of the fit under data splits.

Test 10 ? cross_validation [PASS]

What this test verifies

Result

Test 10 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 10 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:56:52 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 11 ? anderson_darling [FAIL]

Tail-sensitive GOF vs GUE surmise ? full range + height-resolved asymptotic verdict (cf. Test 5)

What this test verifies
What this test verifies

Test 11 [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 11 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-14 23:56:52 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Tail-sensitive GOF vs GUE surmise ? full range + height-resolved asymptotic verdict (cf. Test 5)

Test 11 ? anderson_darling [PASS]

Method: tail-sensitive GOF vs GUE surmise ? full range + height-resolved asymptotic verdict (cf. Test 5)

Test 11 [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

What this test verifies

Result

Plots

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: julia ab_cloud_v23.jl --test 11 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: FAIL | Generated: 2026-09-14 23:56:59 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 12 ? two_sample_ks [FAIL]

Two-sample KS: data vs simulated GUE matrix spectrum.

What this test verifies

METHOD: Two-sample KS: data vs simulated GUE matrix spectrum.

Test 12 [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 12 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: FAIL | Generated: 2026-09-14 23:56:59 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Two-sample KS: data vs simulated GUE matrix spectrum.

Test 12 ? two_sample_ks [FAIL]

METHOD: Two-sample KS: data vs simulated GUE matrix spectrum.

Test 12 [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

What this test verifies

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Result

Plots

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 12 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Test 13 ? number_variance [PASS]

Verdict: PASS | Generated: 2026-09-14 23:57:04 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies ~ (1/??)lnL for GUE ? asymptotic curve comparison.

METHOD: Long-range statistics: ??(L) ~ (1/??)lnL for GUE ? asymptotic curve comparison.

Test 13 [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Result

- ‘plots/plot_01.png’ ? 1600×1000, ABPlotV23 engine (supersampled)

Plots/animation.gif ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 13 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

Independent verification guide

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:57:04 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Long-range statistics: ??(L) ~ (1/??)lnL for GUE ? asymptotic curve comparison.

Test 13 ? number_variance [PASS]

METHOD: Long-range statistics: ??(L) ~ (1/??)lnL for GUE ? asymptotic curve comparison.

What this test verifies

Test 13 [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

- ‘plots/plot_01.png’ ? 1600×1000, ABPlotV23 engine (supersampled)

Plots/animation.gif ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 13 --no-two-pass`

Plots

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Independent verification guide

Test 14 ? spectral_rigidity [PASS]

Verdict: PASS | Generated: 2026-09-14 23:57:12 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

??(L) long-range rigidity vs GUE universal prediction.

METHOD: ??(L) long-range rigidity vs GUE universal prediction.

Result

Test 14 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 14 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:57:12 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 14 ? spectral_rigidity [PASS]

??(L) long-range rigidity vs GUE universal prediction.

METHOD: ??(L) long-range rigidity vs GUE universal prediction.

What this test verifies

Test 14 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Result

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 14 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-14 23:57:32 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 15 ? ab_construction [PASS]

What this test verifies

Mixed: Hermiticity/?_TRB are EXACT (machine-?), ?r? check is STATISTICAL.

METHOD: Mixed: Hermiticity/?_TRB are EXACT (machine-?), ?r? check is STATISTICAL.

Test 15 HARDCORE: full plaquette scan 96x96 (—quartet), signed flux 4/4, empty 9021/9021, worst empty flux 0.00e+00.

Result

Test 15b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Plots/plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 15 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-14 23:57:32 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 15 ? ab_construction [PASS]

Mixed: Hermiticity/?_TRB are EXACT (machine-?), ?r? check is STATISTICAL.

METHOD: Mixed: Hermiticity/?_TRB are EXACT (machine-?), ?r? check is STATISTICAL.

What this test verifies

Test 15 HARDCORE: full plaquette scan 96x96 (—quartet), signed flux 4/4, empty 9021/9021, worst empty flux 0.00e+00.

Test 15b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

- 'plots/plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Result

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 15 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 00:24:43 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 16 ? ab_gue_class [PASS]

What this test verifies

Statistical: ρ ? over the central bulk window vs GUE 0.5992, MC p-value as diagnostics.

METHOD: Statistical: ρ ? over the central bulk window vs GUE 0.5992, MC p-value as diagnostics.

Test 16 HARDCORE pass 2: 96x96, n=5 realizations at $\rho=0.5000$, $N_v=2$, $q=1.000$, $W_{\text{eff}}=1.00$, ensemble deep checks (bootstrap CI, scatter, sign count).

Test 16 HARDCORE: $\rho=0.5798-0.0037$ (n=5), bootstrap CI [0.5732,0.5859] misses GUE, scatter $\rho=0.0082$.

Result

Test 16b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Plots
- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 16 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 00:24:43 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 16 ? ab_gue_class [PASS]

Statistical: ρ ? over the central bulk window vs GUE 0.5992, MC p-value as diagnostics.

METHOD: Statistical: ρ ? over the central bulk window vs GUE 0.5992, MC p-value as diagnostics.

Test 16 HARDCORE pass 2: 96x96, n=5 realizations at $\rho=0.5000$, $N_v=2$, $q=1.000$, $W_{\text{eff}}=1.00$, ensemble deep checks (bootstrap CI, scatter, sign count).

What this test verifies

Test 16 HARDCORE: $\rho=0.5798-0.0037$ (n=5), bootstrap CI [0.5732,0.5859] misses GUE, scatter $\rho=0.0082$.

Test 16b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Result
- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Independent verification guide

Reproduce with: julia ab_cloud_v23.jl --test 16 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Test 17 ? connes_self_duality [PASS]

Verdict: PASS | Generated: 2026-09-15 00:55:45 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies ? ? 1/? to machine precision.

METHOD: EXACT: isospectrality under ? ? 1/? to machine precision.

Test 17 HARDCORE L=24: zero_modes(?=1/2)=4, spectral defects 1.42e-15/1.66e-15, duality defect 3.62e-01.

Test 17 HARDCORE L=48: zero_modes(?=1/2)=4, spectral defects 2.90e-15/2.41e-15, duality defect 3.65e-01.

Test 17 HARDCORE L=96: zero_modes(?=1/2)=4, spectral defects 3.96e-15/3.24e-15, duality defect 3.66e-01.

Test 17b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 17 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 00:55:45 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT: isospectrality under ? ? 1/? to machine precision.

Test 17 ? connes_self_duality [PASS]

METHOD: EXACT: isospectrality under ? ? 1/? to machine precision.

Test 17 HARDCORE L=24: zero_modes(?=1/2)=4, spectral defects 1.42e-15/1.66e-15, duality defect 3.62e-01.

Test 17 HARDCORE L=48: zero_modes(?=1/2)=4, spectral defects 2.90e-15/2.41e-15, duality defect 3.65e-01.

What this test verifies

Test 17 HARDCORE L=96: zero_modes(?=1/2)=4, spectral defects 3.96e-15/3.24e-15, duality defect 3.66e-01.

Result

Test 17b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 17 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 01:01:13 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 18 ? chiral_AIII [PASS]

EXACT: $\|SHS+H\|/\|H\|$? $1e-16$ on the pure lattice ($W=0$); $W>0$ is the honest breaker.

What this test verifies

Test 18 HARDCORE: exact 0.00e+00, ?-row 5.77e-03 (?_rms 0.0058, in window), vortex 0.00e+00 (preserved), S?=I 0.00e+00 (96x96 torus).

Test 18b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 18 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 01:01:13 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT: $\|SHS+H\|/\|H\|$? $1e-16$ on the pure lattice ($W=0$); $W>0$ is the honest breaker.

METHOD: EXACT: $\|SHS+H\|/\|H\|$? $1e-16$ on the pure lattice ($W=0$); $W>0$ is the honest breaker.

Test 18 ? chiral_AIII [PASS]

Test 18 HARDCORE: exact 0.00e+00, ?-row 5.77e-03 (?_rms 0.0058, in window), vortex 0.00e+00 (preserved), S?=I 0.00e+00 (96x96 torus).

What this test verifies

Result

Test 18b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 18 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 01:24:34 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 19 ? dirac_cone [PASS]

EXACT-geometry: zero modes at $(\pi/2,\pi/2)$ require $L/4$??; gap scaling fit R?. FOUR-pass (user spec 2026-09-04): primary fit ? 11-pt hardcore audit ? ?-series ? twist-spectroscopy A

What this test verifies: zero modes at $(\pi/2,\pi/2)$ require $L/4$??; gap scaling fit R?. FOUR-pass (user spec 2026-09-04): primary fit ? 11-pt hardcore audit ? ?-series ? twist-sp

Test 19 AUDIT: v_F _twist = 1.9990 (R? = 1.00000, L=48 y-twist scan), tower 4 @1e-10 with 2+2 polarization, 2?-holonomy 2.5e-17, ?-cusp 6.279, rebuild bit-exact, 1/L? intercept 1

Test 19d [AUDIT pass 4]: 8 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 19 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 01:24:34 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT-geometry: zero modes at $(\pi/2,\pi/2)$ require $L/4$??; gap scaling fit R?. FOUR-pass (user spec 2026-09-04): primary fit ? 11-pt hardcore audit ? ?-series ? twist-spectroscopy A

METHOD: EXACT-geometry: zero modes at $(\pi/2,\pi/2)$ require $L/4$??; gap scaling fit R?. FOUR-pass (user spec 2026-09-04): primary fit ? 11-pt hardcore audit ? ?-series ? twist-sp

Test 19 ? dirac_cone [PASS]

Test 19 AUDIT: v_F _twist = 1.9990 (R? = 1.00000, L=48 y-twist scan), tower 4 @1e-10 with 2+2 polarization, 2?-holonomy 2.5e-17, ?-cusp 6.279, rebuild bit-exact, 1/L? intercept 1

What this test verifies

Result

Test 19d [AUDIT pass 4]: 8 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 19 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 01:24:34 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 19 ? dirac_cone [PASS]

EXACT-geometry: zero modes at $(\pi/2,\pi/2)$ require $L/4??$; gap scaling fit $R?$. FOUR-pass (user spec 2026-09-04): primary fit ? 11-pt hardcore audit ? ?-series ? twist-spectroscopy A

METHOD: EXACT-geometry: zero modes at $(\pi/2,\pi/2)$ require $L/4??$; gap scaling fit $R?$. FOUR-pass (user spec 2026-09-04): primary fit ? 11-pt hardcore audit ? ?-series ? twist-spectroscopy A

What this test verifies

Test 19 AUDIT: v_F _twist = 1.9990 ($R? = 1.00000$, $L=48$ y-twist scan), tower 4 @ $1e-10$ with 2+2 polarization, $2?-holonomy$ $2.5e-17$, $?-cusp$ 6.279, rebuild bit-exact, $1/L?$ intercept 1

Test 19d [AUDIT pass 4]: 8 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 19 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 01:24:34 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 19 ? dirac_cone [PASS]

What this test verifies

EXACT-geometry: zero modes at $(\pi/2,\pi/2)$ require $L/4\in\mathbb{Z}$; gap scaling fit $R\in\mathbb{Z}$. FOUR-pass (user spec 2026-09-04): primary fit $\in\mathbb{Z}$ 11-pt hardcore audit $\in\mathbb{Z}$ π -series $\in\mathbb{Z}$ twist-spectroscopy $A\in\mathbb{Z}$

METHOD: EXACT-geometry: zero modes at $(\pi/2,\pi/2)$ require $L/4\in\mathbb{Z}$; gap scaling fit $R\in\mathbb{Z}$. FOUR-pass (user spec 2026-09-04): primary fit $\in\mathbb{Z}$ 11-pt hardcore audit $\in\mathbb{Z}$ π -series $\in\mathbb{Z}$ twist-spectroscopy $A\in\mathbb{Z}$

Test 19 AUDIT: $v_F \text{ twist} = 1.9990$ ($R\in\mathbb{Z} = 1.00000$, $L=48$ y-twist scan), tower 4 @ $1e-10$ with 2+2 polarization, 2π -holonomy $2.5e-17$, π -cusp 6.279, rebuild bit-exact, $1/L\in\mathbb{Z}$ intercept 1

Result

Test 19d [AUDIT pass 4]: 8 sub-checks, 0 failed $\in\mathbb{Z}$ PASS

Plots/plots/plot_01.png' $\in\mathbb{Z}$ 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_02.png' $\in\mathbb{Z}$ 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_03.png' $\in\mathbb{Z}$ 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' $\in\mathbb{Z}$ animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 19 --no-two-pass`

Independent verification guide
- `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.
(computation log + console capture of this test $\in\mathbb{Z}$ `logs/full_run_log.txt` $\in\mathbb{Z}$ the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 01:25:02 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 20 $\in\mathbb{Z}$ chern_tknn [PASS]

EXACT: FHS $C\in\mathbb{Z}=+1$ on the MAGNETIC BZ ($q\in\mathbb{Z}$ Harper; anchors $\in\mathbb{Z}=1/4,1/3,1/5$); $\in\mathbb{Z}=1/2$ = Dirac touching, per-band C undefined $\in\mathbb{Z}$ touching-ladder rungs need $L\in\mathbb{Z}\in\mathbb{Z}0\pmod4$ so $(\pi/2,\pi/2)\in\mathbb{Z}$ is a Dirac point

METHOD: EXACT: FHS $C\in\mathbb{Z}=+1$ on the MAGNETIC BZ ($q\in\mathbb{Z}$ Harper; anchors $\in\mathbb{Z}=1/4,1/3,1/5$); $\in\mathbb{Z}=1/2$ = Dirac touching, per-band C undefined $\in\mathbb{Z}$ touching-ladder rungs need $L\in\mathbb{Z}\in\mathbb{Z}0\pmod4$ so $(\pi/2,\pi/2)\in\mathbb{Z}$ is a Dirac point

What this test verifies
- 20 HARDCORE verifies $1/4,1/3,1/5\in\mathbb{Z}\in\mathbb{Z}+1\in\mathbb{Z}$; ladder $12..72\in\mathbb{Z}\in\mathbb{Z}1,1,1,1,1,1$; $\in\mathbb{Z}=1/2$ touching $\in\mathbb{Z}L$, mass branch $0\in\mathbb{Z}$.

Test 20b [HARDCORE pass 2]: 3 sub-checks, 0 failed $\in\mathbb{Z}$ PASS

- 'plots/plot_01.png' $\in\mathbb{Z}$ 1600x1000, ABPlotV23 engine (supersampled)

Result
- 'plots/plot_02.png' $\in\mathbb{Z}$ 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_03.png' $\in\mathbb{Z}$ 1600x1000, ABPlotV23 engine (supersampled)

Plots
- 'plots/plot_04.png' $\in\mathbb{Z}$ 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_05.png' $\in\mathbb{Z}$ 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' $\in\mathbb{Z}$ animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 20 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Test 20 ? chern_tknn [PASS]

Verdict: PASS | Generated: 2026-09-15 01:25:02 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

What this test verifies: MAGNETIC BZ (q×q Harper; anchors ?=1/4,1/3,1/5); ?=1/2 = Dirac touching, per-band C undefined ? touching-ladder rungs need L ? 0 (mod 4) so (?/2,

METHOD: EXACT: FHS C?+=+1 on the MAGNETIC BZ (q×q Harper; anchors ?=1/4,1/3,1/5); ?=1/2 = Dirac touching, per-band C undefined ? touching-ladder rungs need L ? 0 (mod

Test 20 HARDCORE: anchors 1/4,1/3,1/5 ? +1 ?; ladder 12..72 ? 1,1,1,1,1,1; ?=1/2 touching ?L, mass branch 0 ?.

Test 20b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)

Plots
- 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)

- 'plots/plot_03.png' ? 1600×1000, ABPlotV23 engine (supersampled)

- 'plots/plot_04.png' ? 1600×1000, ABPlotV23 engine (supersampled)

- 'plots/plot_05.png' ? 1600×1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 20 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 01:25:31 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT geometric phase check.

Test 21 ? gamma_phase [PASS]

METHOD: EXACT geometric phase check

Test 21 HARDCORE: 256-bit a_C err 1.23e-19, b_C err 6.39e-17, arg dev 0.1259°, ?–1e-6 worst dev 0.1259°.

What this test verifies

Result

Test 21b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 21 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 01:25:31 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 21 ? gamma_phase [PASS]

EXACT geometric phase check.

What this test verifies

Test 21 HARDCORE: 256-bit a_C err 1.23e-19, b_C err 6.39e-17, arg dev 0.1259°, ?–1e-6 worst dev 0.1259°.

Test 21b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 21 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 01:25:42 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT flux-phase identity.

METHOD: EXACT flux-phase identity.

Test 22 ? ab_phase [PASS]

Test 22 HARDCORE: 256-bit rational identity true, Float64 err 1.75e-17, flux linearity p=1..3 true.

What this test verifies

Result

Test 22b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 22 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 01:25:42 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 22 ? ab_phase [PASS]

EXACT flux-phase identity.

What this test verifies

Test 22 HARDCORE: 256-bit rational identity true, Float64 err 1.75e-17, flux linearity p=1..3 true.

Test 22b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 22 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 01:25:52 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Log-periodic factor ? semi-statistical curve comparison.

METHOD: Log-periodic factor ? semi-statistical curve comparison.

Test 23 ? fractal_factor [PASS]

Test 23 HARDCORE: |CF|?1 = 0.00e+00 (256-bit), ? err 3.04e-16 (Float64) / 3.28e-05 (LaTeX), c_AB err 6.82e-06.

What this test verifies

Result

Test 23b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 23 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 01:25:52 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 23 ? fractal_factor [PASS]

Log-periodic factor ? semi-statistical curve comparison.

What this test verifies

Test 23 HARDCORE: |CF|?1 = 0.00e+00 (256-bit), ? err 3.04e-16 (Float64) / 3.28e-05 (LaTeX), c_AB err 6.82e-06.

Test 23b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 23 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 01:26:46 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT: plaquette flux = 2?q (mod 2?); tolerance = round-off accumulation across the lattice.

METHOD: EXACT: plaquette flux = 2?q (mod 2?); tolerance = round-off accumulation across the lattice.

Test 24 ? dirac_string_flux [PASS]

Test 24 HARDCORE: 4-config full scan on 96x96, worst empty flux 8.88e-16, hermiticity true.

What this test verifies

Result

Test 24b [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 24 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 01:26:46 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 24 ? dirac_string_flux [PASS]

EXACT: plaquette flux = $2\pi q \pmod{2\pi}$; tolerance = round-off accumulation across the lattice.

What this test verifies: plaquette flux = $2\pi q \pmod{2\pi}$; tolerance = round-off accumulation across the lattice.

Test 24 HARDCORE: 4-config full scan on 96x96, worst empty flux 8.88e-16, hermiticity true.

Test 24b [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 24 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 02:34:35 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT: spectra identical for q and $q+1$ ($\sim 1e-15$); fractional q must DIFFER (non-neutral config).

METHOD: EXACT: spectra identical for q and $q+1$ ($\sim 1e-15$); fractional q must DIFFER (non-neutral config).

Test 25 ? byers_yang [PASS]

Test 25 HARDCORE: Byers-Yang $< 1e-10$ for 3 layouts $\times q \in \{+1, +2, ?1\}$ on 96x96 ? true; fractional $q \in \{0.3, 0.5\} \Rightarrow 1e-3$? true.

What this test verifies

Result

Test 25b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 25 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 02:34:35 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 25 ? byers_yang [PASS]

EXACT: spectra identical for q and q+1 (?~1e-15); fractional q must DIFFER (non-neutral config).

What this test verifies: spectra identical for q and q+1 (?~1e-15); fractional q must DIFFER (non-neutral config).

Test 25 HARDCORE: Byers-Yang ?<1e-10 for 3 layouts xq?{+1,+2,?1} on 96x96 ? true; fractional q?{0.3,0.5} ?>1e-3 ? true.

Test 25b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 25 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 03:07:01 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Mixed: flux identities exact; ?r? on 4 neutral vortices over the bulk window (statistical).

METHOD: Mixed: flux identities exact; ?r? on 4 neutral vortices over the bulk window (statistical).

Test 26 ? pbc_torus [PASS]

Test 26 HARDCORE pass 2: 96x96 torus, exhaustive artifact classification, n=5 ?r? realizations at W_eff=1.00.

Test 26 HARDCORE: artifacts 9024 clean/0 wrap/0 unexplained; ?r?=0.5964–0.0021, scatter ?=0.0046.

What this test verifies

Result

Test 26b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 26 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 03:07:01 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 26 ? pbc_torus [PASS]

Mixed: flux identities exact; ?r? on 4 neutral vortices over the bulk window (statistical).

What this test verifies: flux identities exact; ?r? on 4 neutral vortices over the bulk window (statistical).

Test 26 HARDCORE pass 2: 96x96 torus, exhaustive artifact classification, n=5 ?r? realizations at W_eff=1.00.

Test 26 HARDCORE: artifacts 9024 clean/0 wrap/0 unexplained; ?r?=0.5964–0.0021, scatter ?=0.0046.

Test 26b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 26 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 03:09:45 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT: defect 0 on pure lattice; W>0 row is the honest breaker control. ?-dictionary (v23.2 rework): pass 1 embeds 20,000 zeros (honest ?r? verdict), HARDCORE embeds the full

METHOD: EXACT: defect 0 on pure lattice; W≥0 row is the honest breaker control. ?-dictionary (v23.2 rework): pass 1 embeds 20,000 zeros (honest ?r? verdict), HARDCORE em

Test 27 ? binary_chiral [FAIL]

Test 27 ?-DEEP (50k): pairing exact on the 100000-dim dictionary, ?r? 0.6119 (GUE 0.5996), Poisson control 0.3864, block spread 0.0050, KS D_GUE 0.0216 vs D_GOE 0.0881.

Test 27 HARDCORE: exact 0.00e+00, ?-row 5.77e-03 (?_rms 0.0058), vortex 0.00e+00 (preserved), Coulomb W-scan 0.101?0.746 (ratio 7.42), ?-floor W-independent.

What this test verifies

Result

Test 27b [HARDCORE pass 2]: 8 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 27 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 03:09:45 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 27 ? binary_chiral [PASS]

EXACT: defect 0 on pure lattice; W>0 row is the honest breaker control. ?-dictionary (v23.2 rework): pass 1 embeds 20,000 zeros (honest ?r? verdict), HARDCORE embeds the full

METHOD: EXACT: defect 0 on pure lattice; W>0 row is the honest breaker control. ?-dictionary (v23.2 rework): pass 1 embeds 20,000 zeros (honest ?r? verdict), HARDCORE em

What this test verifies

Test 27 ?-DEEP (50k): pairing exact on the 100000-dim dictionary, ?r? 0.6119 (GUE 0.5996), Poisson control 0.3864, block spread 0.0050, KS D_GUE 0.0216 vs D_GOE 0.0881.

Test 27 HARDCORE: exact 0.00e+00, ?-row 5.77e-03 (?_rms 0.0058), vortex 0.00e+00 (preserved), Coulomb W-scan 0.101?0.746 (ratio 7.42), ?-floor W-independent.

Test 27b [HARDCORE pass 2]: 8 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 27 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Test 28 ? f_gue_merit [FAIL]

Verdict: FAIL | Generated: 2026-09-15 03:33:19 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Statistical: fraction of spectral diagnostics consistent with GUE.

METHOD: Statistical: fraction of spectral diagnostics consistent with GUE.

Test 28 HARDCORE pass 2: 96x96, Nv=4, q=-1.0, W=1.0, n=5 realizations, per-realization f_GUE ensemble.

Test 28 HARDCORE: ?f_GUE?=0.8658-0.0217 (n=5), CI [0.8284,0.9056], scatter ?=0.0485.

Test 28b [HARDCORE pass 2]: 2 sub-checks, 1 failed ? WARN

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 28 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: FAIL | Generated: 2026-09-15 03:33:19 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 28 ? f_gue_merit [FAIL]

Statistical: fraction of spectral diagnostics consistent with GUE.

METHOD: Statistical: fraction of spectral diagnostics consistent with GUE.

Test 28 HARDCORE pass 2: 96x96, Nv=4, q=-1.0, W=1.0, n=5 realizations, per-realization f_GUE ensemble.

What this test verifies

Test 28 HARDCORE: ?f_GUE?=0.8658-0.0217 (n=5), CI [0.8284,0.9056], scatter ?=0.0485.

Test 28b [HARDCORE pass 2]: 2 sub-checks, 1 failed ? WARN

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Result

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 28 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Test 29 ? **dirac_dip** [PASS]

Verdict: PASS | Generated: 2026-09-15 04:28:22 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

METHOD: Semi-exact: DOS dip depth vs prediction.

Test 29 SERIES: 9-pt fine scan @96? (strict dip true, ?_dip=0.0195), contrast(w) 288.00?3.97 (w=0.1?0.8), size contrast 13.33?10.67x

Test 29c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_04.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 29 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 04:28:22 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Semi-exact: DOS dip depth vs prediction.

Test 29 ? **dirac_dip** [FAIL]

METHOD: Semi-exact: DOS dip depth vs prediction.

Test 29 SERIES: 9-pt fine scan @96? (strict dip true, ?_dip=0.0195), contrast(w) 288.00?3.97 (w=0.1?0.8), size contrast 13.33?10.67x

What this test verifies

Test 29c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS

Result

Plots

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_04.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 29 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.
(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 04:28:22 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 29 ? dirac_dip [PASS]

Semi-exact: DOS dip depth vs prediction.

METHOD: Semi-exact DOS dip depth vs prediction.

What this test verifies

Test 29 SERIES: 9-pt fine scan @96? (strict dip true, ?_dip=0.0195), contrast(w) 288.00?3.97 (w=0.1?0.8), size contrast 13.33?10.67x

Test 29c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_04.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 29 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.
(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 04:40:07 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 30 ? vf_scaling [PASS]

What this test verifies

EXACT-geometry: FIRST DISTINCT Dirac level above the zero tower ~ 1/L (the old $ev[j_{0+1}] - ev[j_0]$ hit the valley-degenerate partner ~1e-15 ? gaps 0.0); L/4?? grid up to t31_lmax;

METHOD: EXACT-geometry: FIRST DISTINCT Dirac level above the zero tower ~ 1/L (the old $ev[j_{0+1}] - ev[j_0]$ hit the valley-degenerate partner ~1e-15 ? gaps 0.0); L/4?? grid up to

Test 30 HARDCORE: 11-point fit $R?=0.9999$, $v_F=1.9447$, $E? \bullet L$ $cv=0.75\%$, zero tower 4 ?L.

Result

Test 30b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

- Plots
- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 30 --no-two-pass`

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 04:40:07 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 30 ? vf_scaling [PASS]

EXACT-geometry: FIRST DISTINCT Dirac level above the zero tower ~ 1/L (the old $ev[j_{0+1}] - ev[j_0]$ hit the valley-degenerate partner ~1e-15 ? gaps 0.0); L/4?? grid up to t31_lmax;

METHOD: EXACT-geometry: FIRST DISTINCT Dirac level above the zero tower ~ 1/L (the old $ev[j_{0+1}] - ev[j_0]$ hit the valley-degenerate partner ~1e-15 ? gaps 0.0); L/4?? grid up to

Test 30 HARDCORE: 11-point fit $R?=0.9999$, $v_F=1.9447$, $E? \bullet L$ $cv=0.75\%$, zero tower 4 ?L.

What this test verifies

Test 30b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

- Plots
- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 30 --no-two-pass`

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Test 31 ? hatano_nelson [PASS]

Verdict: PASS | Generated: 2026-09-15 05:29:51 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Deterministic non-Hermitian spectral property: complex eigenvalue count/pattern.

METHOD: Deterministic non-Hermitian spectral property: complex eigenvalue count/pattern.

Test 31 SERIES: n=5 realizations ×??{0.5,0.6,0.7,0.8} @48x48, n_cx mean 0,2225,2281,2295, ?r? 0.8953?0.9253, max|Im| 0.3184?0.9576.

Result Test 31c [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS

- Plots
- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_03.png' ? 1600×1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_04.png' ? 1600×1000, ABPlotV23 engine (supersampled)
 - 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 31 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 05:29:51 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 31 ? hatano_nelson [PASS]

METHOD: Deterministic non-Hermitian spectral property: complex eigenvalue count/pattern.

Test 31 SERIES: n=5 realizations ×??{0.5,0.6,0.7,0.8} @48x48, n_cx mean 0,2225,2281,2295, ?r? 0.8953?0.9253, max|Im| 0.3184?0.9576.

What this test verifies

Test 31c [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS

- Result
- Plots
- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_02.png' ? 1600×1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_03.png' ? 1600×1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_04.png' ? 1600×1000, ABPlotV23 engine (supersampled)
 - 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: julia ab_cloud_v23.jl --test 31 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Test 31 ? hatano_nelson [PASS]

Verdict: PASS | Generated: 2026-09-15 05:29:51 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Why this test verifies spectral property: complex eigenvalue count/pattern.

METHOD: Deterministic non-Hermitian spectral property: complex eigenvalue count/pattern.

Test 31 SERIES: n=5 realizations ×??{0.5,0.6,0.7,0.8} @48x48, n_cx mean 0,2225,2281,2295, ?r? 0.8953?0.9253, max|Im| 0.3184?0.9576.

Test 31c [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_04.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 31 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 06:47:18 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Statistical: ?r? = 0.5992 – 0.022 (tolerance band = monograph §11.1); CI printed as diagnostics. Effective disorder capped: W_eff = min(ab_W, ab_W_max ? 1.0) ? disorder must not

Test 32 ? rmean_bootstrap [PASS]

Method: Statistical: r = 0.5992 – 0.022 (tolerance band = monograph §11.1); CI printed as diagnostics. Effective disorder capped: W_eff = min(ab_W, ab_W_max ? 1.0) ? disorder must not

Test 32 HARDCORE pass 2: 96x96, n?5 per row, comparison-row Nv=256 (anchor density), ensemble deep checks (bootstrap CI, scatter, sign count).

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

What this test verifies

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik} \bullet N_{\cdot,} + 1), \text{ hopping } t = 1.$$

The ratio (disorder energy q•W) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

? q•W ? t (e.g. W=4, q=0.3 ? V?1.2 next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the eigenstates, states localize on the potential relief, and ?r? drifts from GUE (0.5992) toward Poisson (0.3863) by an essentially random amount. A strongly disordered, spinning vortex system can therefore emit ALMOST ANY statistics ? the numbers depend on the effective rotation speed (W/t) and on the twisting/vortex arrangement (Nv, layout, charges) of that realization, not on the AB physics.

? q•W ? t (W ? 1): the Aharonov-Bohm phases dominate the hopping, states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at W_eff = min(cfg.ab_W, W_max = 1.0): they certify the PHASE physics, not the disorder physics.

Effective disorder: W_eff = 1.00 = min(ab_W=4.00, ab_W_max=1.00); disorder energy q•W_eff = 1.00 vs hopping t = 1.0

Comparison row q=0.300: W_row=0.50, Nv_row=256 ? ?r?=0.6012 – 0.0015; nearest class GUE (d_GOE=0.0705, d_GUE=0.0016, d_Poi=0.2149); vortex density 2.78% vs monograph

Test 32 HARDCORE row q=1.000: n=5, bootstrap CI [0.5838, 0.5892] misses GUE; scatter ?=0.0034 (within band).

Test 32 HARDCORE row q=0.300: n=5, bootstrap CI [0.5999, 0.6026] misses GUE; scatter ?=0.0017 (within band).

Test 32b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 32 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 06:47:18 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Statistical: ?r? = 0.5992 – 0.022 (tolerance band = monograph §11.1); CI printed as diagnostics. Effective disorder capped: W_eff = min(ab_W, ab_W_max ? 1.0) ? disorder must not

METHOD: Statistical: ?r? = 0.5992 – 0.022 (tolerance band = monograph §11.1); CI printed as diagnostics. Effective disorder capped: W_eff = min(ab_W, ab_W_max ? 1.0) ? disorder must not

Test 32 ? rmean_bootstrap [PASS]

Test 32 HARDCORE pass 2: 96x96, n?5 per row, comparison-row Nv=256 (anchor density), ensemble deep checks (bootstrap CI, scatter, sign count).

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

What this test verifies

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k \cdot W / (r_{ik} \cdot N_{\cdot} + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy q•W) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

$\phi \propto W \propto t$ (e.g. $W=4$, $\phi=0.3 \propto V \propto 1.2$ next to a core): the disorder landscape dominates the vortex motion $\propto$ vortices pin/steer the eigenstates, states localize on the potential relief, and ϕ drifts from GUE (0.5992) toward Poisson (0.3863) by an essentially random amount. A strongly disordered, spinning vortex system can therefore emit ALMOST ANY statistics $\propto$ the numbers depend on the effective rotation speed (W/t) and on the twisting/vortex arrangement (N_v , layout, charges) of that realization, not on the AB physics.

$\phi \propto W \propto t$ ($W \propto 1$): the Aharonov-Bohm phases dominate the hopping, states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$: they certify the PHASE physics, not the disorder physics.

Effective disorder: $W_{\text{eff}} = 1.00 = \min(\text{ab_}W=4.00, \text{ab_}W_{\text{max}}=1.00)$; disorder energy $\phi \propto W_{\text{eff}} = 1.00$ vs hopping $t = 1.0$

Comparison row $\phi=0.300$: $W_{\text{row}}=0.50$, $N_v_{\text{row}}=256 \propto \phi_{\text{r}}=0.6012 - 0.0015$; nearest class GUE ($d_{\text{GOE}}=0.0705$, $d_{\text{GUE}}=0.0016$, $d_{\text{Poi}}=0.2149$); vortex density 2.78% vs monograph

Test 32 HARDCORE row $\phi=1.000$: $n=5$, bootstrap CI [0.5838, 0.5892] misses GUE; scatter $\phi=0.0034$ (within band).

Test 32 HARDCORE row $\phi=0.300$: $n=5$, bootstrap CI [0.5999, 0.6026] misses GUE; scatter $\phi=0.0017$ (within band).

Test 32b [HARDCORE pass 2]: 2 sub-checks, 0 failed $\propto$ PASS

Result - 'plots/plot_01.png' $\propto$ 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' $\propto$ animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 32 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test $\propto$ logs/full_run_log.txt $\propto$ the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 07:26:31 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Statistical finite-size trend: monotone approach to 0.594-0.599; monograph anchors $L \propto 50$, extended points $L=80/112$ (user spec 2026-08-29) check the GUE plateau; W_{eff} capped

METHOD: Statistical finite-size trend: monotone approach to 0.594-0.599; monograph anchors $L \propto 50$, extended points $L=80/112$ (user spec 2026-08-29) check the GUE plateau; W_{eff}

Test 33 $\propto$ L_scaling_rmean [PASS]

Test 33 HARDCORE pass 2: $n \propto 5$ per L , weighted GUE-plateau ?? over $L \propto 50$, L capped at 112 (RAM honesty).

WHY W (DISORDER) IS CAPPED AT $W_{\text{max}} \propto 1.0 \propto$ disorder $\propto$ vortex motion $\propto$ statistics:

What this test verifies: potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik} \sum N_{i,j} + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \bullet W$) / (kinetic energy t) decides what the vortex cloud does and which spectral statistics come out:

$\sum q \bullet W \gg t$ (e.g. $W=4$, $q=0.3 \Rightarrow V \approx 1.2$ next to a core): the disorder landscape dominates the vortex motion $\Rightarrow$ vortices pin/steer the eigenstates, states localize on the potential relief, and $\langle r \rangle$ drifts from GUE (0.5992) toward Poisson (0.3863) by an essentially random amount. A strongly disordered, spinning vortex system can therefore emit ALMOST ANY statistics $\Rightarrow$ the numbers depend on the effective rotation speed (W/t) and on the twisting/vortex arrangement (N_v , layout, charges) of that realization, not on the AB physics.

$\sum q \bullet W \lesssim t$ ($W \lesssim 1$): the Aharonov-Bohm phases dominate the hopping, states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$: they certify the PHASE physics, not the disorder physics.

Test 33 RAM gate: $L=112$ skipped (needs ≈ 10.6 GB peak, free ≈ 4.3 GB).

Extended L-scan to $L=80$ (user spec 2026-08-29); $W_{\text{eff}}=1.00$; verdict $q=0.30$; trend $0.434 \approx 0.556 \approx 0.593 \approx 0.6 \approx 0.6$

Test 33 HARDCORE plateau ($q=0.30$): weighted $\langle r \rangle = 0.6004$ vs GUE ($|\langle r \rangle| = 0.0012$, in band), $\langle \langle r \rangle \rangle / \text{dof} = 0.01$.

Test 33b (hardcore): plateau weighted $\langle r \rangle = 0.6004$, $\langle \langle r \rangle \rangle / \text{dof} = 0.01 \Rightarrow$ PASS

'plots/plot_01.png' $\approx 1600 \times 1000$, ABPlotV23 engine (supersampled)

Result

- 'plots/animation.gif' $\Rightarrow$ animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 33 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test $\Rightarrow$ logs/full_run_log.txt $\Rightarrow$ the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 07:26:31 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 33 $\approx$ I_scaling_rmean [PASS]

What this test verifies

Statistical finite-size trend: monotone approach to 0.594-0.599; monograph anchors L?50, extended points L=80/112 (user spec 2026-08-29) check the GUE plateau; W_eff capped

METHOD: Statistical finite-size trend: monotone approach to 0.594-0.599; monograph anchors L?50, extended points L=80/112 (user spec 2026-08-29) check the GUE plateau; W

Test 33 HARDCORE pass 2: n ? 5 per L, weighted GUE-plateau ?? over L ? 50, L capped at 112 (RAM honesty).

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik}^2 + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \bullet W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

$q \bullet W \gg t$ (e.g. $W=4$, $q=0.3 \Rightarrow V \approx 1.2$ next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and r drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (N_v ,

layout, charges) of that realization, not on the AB physics.

$q \bullet W \ll t$ ($W \ll 1$): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$:

they certify the PHASE physics, not the disorder physics.

Test 33 RAM gate: L=112 skipped (needs ? 10.6 GB peak, free ? 4.3 GB).

Extended L-scan to L=80 (user spec 2026-08-29); $W_{\text{eff}}=1.00$; verdict $q=0.30$; trend $0.434 \rightarrow 0.556 \rightarrow 0.593 \rightarrow 0.6 \rightarrow 0.6$

Test 33 HARDCORE plateau ($q=0.30$): weighted $r=0.6004$ vs GUE ($|r|=0.0012$, in band), $??/\text{dof}=0.01$.

Test 33b (hardcore): plateau weighted $r=0.6004$, $??/\text{dof}=0.01 \Rightarrow \text{PASS}$

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Result

Plots

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 33 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Test 34 ? direct_vs_zeta [PASS]

Verdict: FAIL | Generated: 2026-09-15 07:51:46 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Statistical: unfolded-spacing KS (effect size $D < 0.10$) + Montgomery $R?(s)$ vs ?-zeros and GUE. full 50k range, not ?-5000.

METHOD: Statistical: unfolded-spacing KS (effect size $D < 0.10$) + Montgomery $R?(s)$ vs ?-zeros and GUE. full 50k range, not ?-5000.

Test 34 HARDCORE pass 2: n=5 realizations, $R? ds=0.05/s_max=4.0$, per-realization KS stability.

Test 34 HARDCORE: $D=0.1096$ ($p=2.236e-186$), per-real D med 0.1153 (spread 0.0923..0.1189), $?|?R?|?=0.0002$, $d_GUE=0.8764$.

Test 34b [HARDCORE pass 2]: 4 sub-checks, 1 failed ? WARN

Result

- ‘plots/plot_01.png’ ? 1600×1000, ABPlotV23 engine (supersampled)

- ‘plots/plot_02.png’ ? 1600×1000, ABPlotV23 engine (supersampled)

Plots

- ‘plots/plot_03.png’ ? 1600×1000, ABPlotV23 engine (supersampled)

- ‘plots/animation.gif’ ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 34 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run’s single consolidated log file)

Independent verification guide

Verdict: FAIL | Generated: 2026-09-15 07:51:46 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Statistical: unfolded-spacing KS (effect size $D < 0.10$) + Montgomery $R?(s)$ vs ?-zeros and GUE. full 50k range, not ?-5000.

Test 34 ? direct_vs_zeta [FAIL]

METHOD: Statistical: unfolded-spacing KS (effect size $D < 0.10$) + Montgomery $R?(s)$ vs ?-zeros and GUE. full 50k range, not ?-5000.

Test 34 HARDCORE pass 2: n=5 realizations, $R? ds=0.05/s_max=4.0$, per-realization KS stability.

Test 34 HARDCORE: $D=0.1096$ ($p=2.236e-186$), per-real D med 0.1153 (spread 0.0923..0.1189), $?|?R?|?=0.0002$, $d_GUE=0.8764$.

What this test verifies

Result

Test 34b [HARDCORE pass 2]: 4 sub-checks, 1 failed ? WARN

Plots

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 34 --no-two-pass`

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: FAIL | Generated: 2026-09-15 08:17:49 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 35 ? form_factor_Kt [FAIL]

Statistical: raw $(1/n)|e^{it}|$? vs finite-box GUE reference $K_{\text{box}}(t)=t(1+\text{sinc}(2t))$; RMS over $t\in[0,3]$.

METHOD: Statistical: raw $(1/n)|e^{it}|$? vs finite-box GUE reference $K_{\text{box}}(t)=t(1+\text{sinc}(2t))$; RMS over $t\in[0,3]$.

What this test verifies

Test 35 HARDCORE pass 2: n=5 per-realization $K(t)$, 33-point t-grid, ensemble mean verdict.

Test 35 HARDCORE: ensemble RMS=0.6143, corr_box=0.6379, corr_?=0.8123, RMS scatter ?=0.1856.

Test 35b [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 35 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: FAIL | Generated: 2026-09-15 08:17:49 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 35 ? form_factor_Kt [FAIL]

What this test verifies

Statistical: raw $(1/n)|\psi\rangle^{\dagger}\psi\rangle$ vs finite-box GUE reference $K_{\text{box}}(t)=t(1-\text{sinc}(2t))$; RMS over $t\in[0,0.3]$.

METHOD: Statistical: raw $(1/n)|\psi\rangle^{\dagger}\psi\rangle$ vs finite-box GUE reference $K_{\text{box}}(t)=t(1-\text{sinc}(2t))$; RMS over $t\in[0,0.3]$.

Test 35 HARDCORE pass 2: $n=5$ per-realization $K(t)$, 33-point t -grid, ensemble mean verdict.

Test 35 HARDCORE: ensemble RMS=0.6143, $\text{corr}_{\text{box}}=0.6379$, $\text{corr}_{\text{?}}=0.8123$, RMS scatter $\text{?}=0.1856$.

Result

Test 35b [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

Plots
- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 35 --no-two-pass`

Independent verification guide
- `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(computation log + console capture of this test ? `logs/full_run_log.txt` ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 09:08:55 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 36 ? byte_robust [PASS]

Mixed: $|\psi\rangle$ under 256-level quantization + RNG entropy (statistical). Sample accumulated to 25 000 (pass 1) / 50 000 (pass 2) eigenvalues; verdict = multi-window average over 120

METHOD: Mixed: $|\psi\rangle$ under 256-level quantization + RNG entropy (statistical). Sample accumulated to 25 000 (pass 1) / 50 000 (pass 2) eigenvalues; verdict = multi-window average

What this test verifies

What this test verifies
- `logs/stdout_capture.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik}^2 + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $\sum q_k W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

$\sum q_k W \gg t$ (e.g. $W=4$, $q=0.3 \Rightarrow V \approx 1.2$ next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and ψ drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (N_v ,

layout, charges) of that realization, not on the AB physics.

$\sum q_k W \ll t$ ($W \approx 1$): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_W}, W_{\text{max}} = 1.0)$:

they certify the PHASE physics, not the disorder physics.

Sample: 55290 eigs (pass 2 (secondary), target 50000, reached=yes); $W_{\text{eff}}=1.00$.

Byte robustness verdict = multi-window $|\langle r \rangle|=0.0009$ (tol 0.0200, 120 windows); single-window $|\langle r \rangle|=0.0061$ (diagnostic).

Result

Test 36: byte robust ? $r_{256}=0.5756$, $|\langle r \rangle|_{\text{multi}}=0.0009$ (120 windows), $|\langle r \rangle|_{1\text{win}}=0.0061$ (diag), $H=7.9991$ bits, $\langle z \rangle=-0.234$, $n=55288$? PASS

no plots queued by this pass ? physics series live in the HARDCORE pass-2 report: hc_verdict_table queues a sub-check verdict map there, and the deep-audit tests queue their own plots

Reproduce with: `julia ab_cloud_v23.jl --test 36 --no-two-pass`

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 09:08:55 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 36 ? byte_robust [PASS]

Mixed: $|\langle r \rangle|$ under 256-level quantization + RNG entropy (statistical). Sample accumulated to 25 000 (pass 1) / 50 000 (pass 2) eigenvalues; verdict = multi-window average over 120

METHOD: Mixed: $|\langle r \rangle|$ under 256-level quantization + RNG entropy (statistical). Sample accumulated to 25 000 (pass 1) / 50 000 (pass 2) eigenvalues; verdict = multi-window average

What this test verifies

WHY W (DISORDER) IS CAPPED AT $W_{\text{max}} ? 1.0$? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik}^{2N} + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \bullet W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

? $q \bullet W ? t$ (e.g. $W=4$, $q=0.3 ? V \approx 1.2$ next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and $\langle r \rangle$ drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (N_v ,

layout, charges) of that realization, not on the AB physics.

? $q \bullet W ? t$ ($W ? 1$): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_W}, W_{\text{max}} = 1.0)$:

they certify the PHASE physics, not the disorder physics.

Sample: 55290 eigs (pass 2 (secondary), target 50000, reached=yes); W_eff=1.00.

Byte robustness verdict = multi-window |??r|=0.0009 (tol 0.0200, 120 windows); single-window |?r|=0.0061 (diagnostic).

Result

Test 36: byte robust ? r_256=0.5756, |?r|_multi=0.0009 (120 windows), |?r|_1win=0.0061 (diag), H=7.9991 bits, ??_z=-0.234, n=55288 ? PASS

Plots

(no plots queued by this pass ? physics series live in the HARDCORE pass-2 report: hc_verdict_table queues a sub-check verdict map there, and the deep-audit tests queue their own)

Reproduce with: julia ab_cloud_v23.jl --test 36 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 09:09:07 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 37 ? half_factorial_gamma [PASS]

EXACT: ?(1/2) = ??, (1/2)! = ?(3/2) = ??/2, ladder + recurrence (suite's own Lanczos lgamma); UNIQUENESS of the ?-construction via the reflection self-dual fixed point ?(1/2)? = 1

METHOD: EXACT: ?(1/2) = ??, (1/2)! = ?(3/2) = ??/2, ladder + recurrence (suite's own Lanczos lgamma); UNIQUENESS of the ?-construction via the reflection self-dual fixed point

What this test verifies

Test 37 HARDCORE: 256-bit DE ?(1/2) err 1.20e-28, Float64 round-off ?(1/2) 5.83e-16 / (1/2)! 2.94e-16, N=32/?? err 2.51e-26, ladder/self-dual/recurrence exact.

Test 37b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Result: 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Plots

Reproduce with: julia ab_cloud_v23.jl --test 37 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run's single consolidated log file)

Independent verification guide

Verdict: PASS | Generated: 2026-09-15 09:09:07 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

EXACT: ?(1/2) = ??, (1/2)! = ?(3/2) = ??/2, ladder + recurrence (suite's own Lanczos lgamma); UNIQUENESS of the ?-construction via the reflection self-dual fixed point ?(1/2)? = 1

METHOD: EXACT: ?(1/2) = ??, (1/2)! = ?(3/2) = ??/2, ladder + recurrence (suite's own Lanczos lgamma); UNIQUENESS of the ?-construction via the reflection self-dual fixed point

Test 37 ? half_factorial_gamma [PASS]

Test 37 HARDCORE: 256-bit DE ?(1/2) err 1.20e-28, Float64 round-off ?(1/2) 5.83e-16 / (1/2)! 2.94e-16, N=32/?? err 2.51e-26, ladder/self-dual/recurrence exact.

What this test verifies

Result

Test 37b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Plots

- 'plots/plot_01.png' ? 1600×1000, ABPlotV23 engine (supersampled)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 37 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Verdict: PASS | Generated: 2026-09-15 17:16:11 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

Test 38 ? curie_point [PASS]

STATISTICAL+EXACT (v22.2 three-pass rewrite): order parameter $\langle T \rangle = \langle \ln[p_{\text{GUE}}(s)/p_{\text{GOE}}(s)] \rangle$? ? the KL lean ? on Chebyshev-unfolded spacings of a paired-realization vortex-

What this test verifies: EXACT (v22.2 three-pass rewrite): order parameter $\langle T \rangle = \langle \ln[p_{\text{GUE}}(s)/p_{\text{GOE}}(s)] \rangle$? ? the KL lean ? on Chebyshev-unfolded spacings of a paired-realiza

Test 38 (v23 three-pass) t38: PASS • Curie 3-pass (72?+96?+38c) ? 486.9 min, 346 spectra in cache

Result

- 'plots/t38_curie_panels.png' ? Test 38 ? Curie point (38a + 38b + 38c) (journal-engine multi-panel figure)

Plots

Reproduce with: `julia ab_cloud_v23.jl --test 38 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(computation log + console capture of this test ? logs/full_run_log.txt ? the run’s single consolidated log file)

Independent verification guide